

Practicing

**VulnMapper: Network Device Mapping and
Vulnerability Assessment Tool**

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1. Project Overview

1.1 Vision

The objective of VulnMapper is to automate network reconnaissance and vulnerability assessment tasks. The tool enables users to identify active hosts, enumerate services, discover potential vulnerabilities, locate systems that may be susceptible to weak passwords, and generate organized scan results for further analysis.

1.2 Methodology & Tools

The project was developed using Bash scripting on Kali Linux. The primary tools used include:

- Nmap
- Nmap NSE Vulnerability Scripts
- SearchSploit
- Grep
- Awk
- Zip

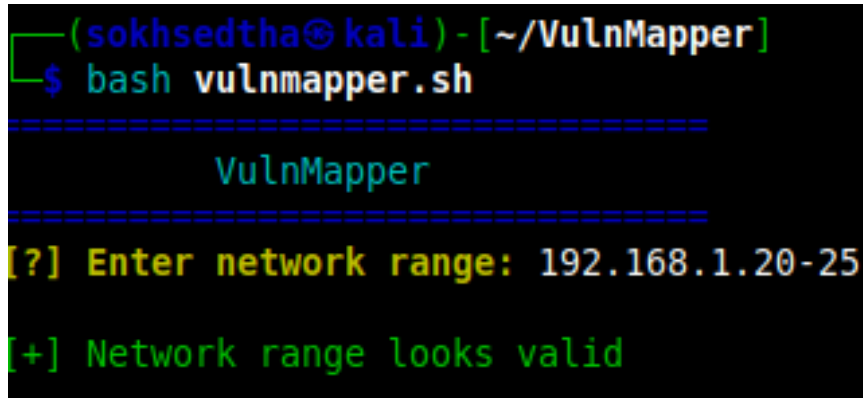
The workflow consists of:

1. User input and validation
2. Host discovery
3. Service enumeration
4. Vulnerability scanning
5. Credential target identification
6. Exploit mapping
7. Post-scan analysis
8. Result archiving

2. System Operational Workflow

2.1 User Input and Pre-Scan Validation

The user specifies a network range to scan.

A terminal window with a black background and colorful text. The prompt is '(sokhsedtha@kali) - [~/VulnMapper]'. The user enters '\$ bash vulnmapper.sh'. The script outputs a blue separator line, then 'VulnMapper' in green, followed by another blue separator line. It then prompts '[?] Enter network range: 192.168.1.20-25' in yellow. Finally, it outputs '[+] Network range looks valid' in green.

```
(sokhsedtha@kali) - [~/VulnMapper]
$ bash vulnmapper.sh
=====
VulnMapper
=====
[?] Enter network range: 192.168.1.20-25
[+] Network range looks valid
```

The script validates the supplied range using Nmap list scan mode. Invalid ranges terminate execution while valid ranges allow the assessment process to continue.

The user also selects either:

- Built-in password list (rockyou.txt)
- Custom password list

```
=====
VulnMapper
=====
[?] Enter network range: 192.168.1.0/24

[+] Network range looks valid

[*] Password List Selection
[1] Built-in password list
[2] Custom password list
Choose option: 2
Enter password list path: /home/sokhsedtha/VulnMapper/password.txt

[+] Password list selected:
/home/sokhsedtha/VulnMapper/password.txt

(sokhsedtha@kali) - [~/VulnMapper]
$ bash vulnmapper.sh
=====
VulnMapper
=====
[?] Enter network range: 192.168.1.0/24

[+] Network range looks valid

[*] Password List Selection
[1] Built-in password list
[2] Custom password list
Choose option: 1

[+] Password list selected:
/usr/share/wordlists/rockyou.txt

(sokhsedtha@kali) - [~/VulnMapper]
```

This ensures flexibility for future credential auditing activities.

3. Technical Audit Results (Basic Scan)

3.1 Service Enumeration

```
[*] Discovering live hosts...
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 10:32 -0400
Nmap scan report for 192.168.1.23
Host is up (0.96s latency).
MAC Address: C8:F7:33:10:98:A2 (Intel Corporate)
Nmap scan report for 192.168.1.22
Host is up.
Nmap done: 6 IP addresses (2 hosts up) scanned in 7.18 seconds

[+] Host discovery completed
[+] Results saved to scans/hosts.txt
```

After identifying active hosts, the script performs service enumeration using:

```
nmap -sV --top-ports 10
```

This phase identifies:

- Open ports
- Running services
- Service versions

Example findings:

- OpenSSH 10.3p1
- Microsoft HTTPAPI 2.0

Results are stored individually for each host inside the services directory.

```
Nmap done: 1 IP address (1 host up) scanned in 3.08 seconds
[+] Scanning 192.168.1.22
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 10:32 -0400
Nmap scan report for 192.168.1.22
Host is up (0.0059s latency).

PORT      STATE      SERVICE      VERSION
21/tcp    filtered  ftp
22/tcp    open      ssh          OpenSSH 10.3p1 Debian 2 (protocol 2.0)
23/tcp    filtered  telnet
25/tcp    filtered  smtp
80/tcp    filtered  http
110/tcp   filtered  pop3
139/tcp   filtered  netbios-ssn
443/tcp   filtered  https
445/tcp   filtered  microsoft-ds
3389/tcp   filtered  ms-wbt-server
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.36 seconds
```

3.2 Credential Audit Verification

The tool automatically searches for services commonly associated with authentication:

- FTP (21)
- SSH (22)
- Telnet (23)
- SMB (445)

Detected hosts are recorded in credential_targets.txt.

The user can select either a built-in or custom password list for future credential auditing operations.

```
(sokhsedtha@kali) - [~/VulnMapper]
$ cat scans/credential_targets.txt
scans/services/192.168.1.23.txt:21/tcp    filtered ftp
scans/services/192.168.1.22.txt:21/tcp    filtered ftp
scans/services/192.168.1.23.txt:22/tcp    filtered ssh
scans/services/192.168.1.22.txt:22/tcp    open  ssh      OpenSSH 10.3p1 Debian 2 (protocol 2.0)
scans/services/192.168.1.23.txt:23/tcp    filtered telnet
scans/services/192.168.1.22.txt:23/tcp    filtered telnet
scans/services/192.168.1.23.txt:445/tcp   filtered microsoft-ds
scans/services/192.168.1.22.txt:445/tcp   filtered microsoft-ds
(sokhsedtha@kali) - [~/VulnMapper]
```

4. Technical Audit Results (Full Scan)

4.1 Advanced Vulnerability Discovery (NSE)

Nmap NSE vulnerability scripts are executed using:

```
nmap --script vuln
```

The scan identifies known vulnerabilities and security weaknesses exposed by discovered services.

Results are stored in vulns.txt.

```
(sokhsedtha@kali)-[~/VulnMapper]
$ grep -A 10 "VULNERABLE" scans/vulns.txt
VULNERABLE:
Slowloris DOS attack
  State: LIKELY VULNERABLE
  IDs: CVE:CVE-2007-6750
  Slowloris tries to keep many connections to the target web server open and hold
  them open as long as possible. It accomplishes this by opening connections to
  the target web server and sending a partial request. By doing so, it starves
  the http server's resources causing Denial Of Service.

  Disclosure date: 2009-09-17
  References:
    https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
    http://ha.ckers.org/slowloris/

VULNERABLE:
SSL POODLE information leak
  State: LIKELY VULNERABLE
  IDs: BID:70574 CVE:CVE-2014-3566
  The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
  products, uses nondeterministic CBC padding, which makes it easier
  for man-in-the-middle attackers to obtain cleartext data via a
  padding-oracle attack, aka the "POODLE" issue.

  Disclosure date: 2014-10-14
  Check results:
    TLS_RSA_WITH_AES_128_CBC_SHA
    TLS_FALLBACK_SCSV properly implemented
  References:

VULNERABLE:
SSL POODLE information leak
  State: LIKELY VULNERABLE
  IDs: BID:70574 CVE:CVE-2014-3566
  The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
  products, uses nondeterministic CBC padding, which makes it easier
  for man-in-the-middle attackers to obtain cleartext data via a
  padding-oracle attack, aka the "POODLE" issue.

  Disclosure date: 2014-10-14
  Check results:
    TLS_RSA_WITH_AES_128_CBC_SHA
    TLS_FALLBACK_SCSV properly implemented
  References:

(sokhsedtha@kali)-[~/VulnMapper]
```

4.2 Automated Exploit Mapping

The script integrates SearchSploit to identify publicly available exploit references related to discovered services.

Example:

searchsploit OpenSSH

Results are stored in exploits.txt.

```
(sokhsedtha@kali) (~/.VulnMapper)
$ cat scans/exploits.txt
```

Exploit Title	Path
Debian OpenSSH - (Authenticated) Remote Sslinux Privilege Escalation	linux/remote/0094.txt
Dropbear / OpenSSH Server - 'MAX_UNAUTH_CLIENTS' Denial of Service	multiple/dos/2572.pl
FreeBSD OpenSSH 3.5p1 - Remote Command Execution	Freebsd/remote/17462.txt
glibc-2.2 / openssh-2.3.0p1 / glibc 2.1.9x - File Read	linux/local/2588.sh
Novell Netware 6.5 - OpenSSH Remote Stack Overflow	novell/dos/14866.txt
OpenSSH 1.2 - '.scp' File Create/Overwrite	linux/remote/20233.sh
OpenSSH 2.3 < 7.7 - Username Enumeration	linux/remote/45233.py
OpenSSH 2.3 < 7.7 - Username Enumeration (PoC)	linux/remote/45210.py
OpenSSH 2.x/3.0.1/3.0.2 - Channel Code Off-by-One	unix/remote/21314.txt
OpenSSH 2.x/3.x - Kerberos 4 GTG/AFS Token Buffer Overflow	linux/remote/21402.txt
OpenSSH 3.x - Challenge-Response Buffer Overflow (1)	unix/remote/21578.txt
OpenSSH 3.x - Challenge-Response Buffer Overflow (2)	unix/remote/21579.txt
OpenSSH 4.3 p1 - Duplicated Block Remote Denial of Service	multiple/dos/2444.sh
OpenSSH 6.8 < 6.9 - 'PTY' Local Privilege Escalation	linux/local/41173.c
OpenSSH 7.2 - Denial of Service	linux/dos/40888.py
OpenSSH 7.2p1 - (Authenticated) xauth Command Injection	multiple/remote/39569.py
OpenSSH 7.2p2 - Username Enumeration	linux/remote/40136.py
OpenSSH < 6.6 SFTP (x64) - Command Execution	linux_x86_64/remote/45000.c
OpenSSH < 6.6 SFTP - Command Execution	linux/remote/45001.py
OpenSSH < 7.4 - 'UsePrivilegeSeparation Disabled' Forwarded Unix Domain Sockets Privilege Escalation	linux/local/40962.txt
OpenSSH < 7.4 - agent Protocol Arbitrary Library Loading	linux/remote/40963.txt
OpenSSH < 7.7 - User Enumeration (2)	linux/remote/45939.py
OpenSSH SCP Client - Write Arbitrary Files	multiple/remote/46516.py
OpenSSH server (sshd) 9.8p1 - Race Condition	linux/remote/52269.c
OpenSSH/PAM 3.6.1p1 - 'gossh.sh' Remote Users Ident	linux/remote/26.sh
OpenSSH/PAM 3.6.1p1 - Remote Users Discovery Tool	linux/remote/25.c
OpenSSH 7.2p2 - Username Enumeration	linux/remote/40113.txt
Portable OpenSSH 3.6.1p-PAM/4.1-SuSE - Timing Attack	multiple/remote/3303.sh

Shellcodes: No Results

5. Data Management & Deliverables

5.1 Post-Scan Search Function

```
[*] Post-Scan Search
Enter IP Address: 192.168.1.22

=====
Host Summary
=====

IP Address: 192.168.1.22

Open Ports:
22/tcp    open      ssh          OpenSSH 10.3p1 Debian 2 (protocol 2.0)

Credential Audit Candidate: YES

Exploit References: YES

Vulnerability Findings: YES
```

The user may search for a specific IP address after scanning.

The tool displays:

- Open ports
- Credential audit status
- Vulnerability findings
- Exploit references

This feature provides a quick summary of discovered hosts.

5.2 Automated Archival

```
[*] Creating archive...  
[+] Archive created  
[+] File: VulnMapper_Results.zip
```

```
(sokhsedtha@kali)-[~/VulnMapper]  
$ ls -lh VulnMapper_Results.zip  
-rw-rw-r-- 1 sokhsedtha sokhsedtha 11K Jun 12 10:34 VulnMapper_Results.zip
```

After scan completion, all collected results are compressed automatically into:

VulnMapper_Results.zip

The archive contains:

- hosts.txt
- live_hosts.txt
- service enumeration results
- vulnerability findings
- credential audit targets
- exploit references

This archive can be shared, backed up, or used for reporting purposes.